Sparse Rationale-Constrained Attention for Hate Speech Detection

Anonymous Authors
Anonymous Institution
anonymous@example.org

Abstract

Hate speech classifiers deployed in moderation pipelines must justify their decisions, yet attention weights are routinely shown to be unfaithful explanations of model behaviour. Sparse Rationale-Constrained Attention (SRA), recently introduced as a remedy, replaces softmax with sparsemax and supervises a single attention head with token-level rationales. We revisit SRA on HateXplain and show that, while the structural insight is correct, the *instantiation* is severely under-constrained: a single supervised head leaks explanatory mass into the remaining eleven through the residual stream. We propose an all-layer, all-head sparsemax projection with mean-squared rationale supervision (C4) and a top-six layer variant (C5). Across five seeds we observe (i) classification parity with the softmax baseline (macro-F1 0.6823 ± 0.0063 vs. 0.6814 ± 0.0086, TOST p=0.047 at $\pm$1.0 pp), (ii) a 111% relative gain in ERASER comprehensiveness AOPC (0.338 vs. 0.160 for SRA, Wilcoxon p=0.031, d=8.36), and (iii) a 47% reduction in sufficiency AOPC, all at zero additional inference cost. We honestly disclose that the pre-registered $2\times$ adversarial-swap robustness threshold (H4) is missed: the observed C4/C2 KL ratio is $1.47\times$. A structural argument (Proposition 1) and a layerwise ablation explain why supervising every head is necessary and sufficient: it removes the residual escape route that the original SRA inadvertently leaves open.

1 Introduction

Content-moderation systems are increasingly required, by regulation and by users, to *explain* why a post was removed or flagged. Attention weights are the most convenient candidate explanation in transformer classifiers, but a now-classical line of work has shown them to be only loosely correlated with model behaviour [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019]. The community has reacted along two axes: replace softmax with a sparse alternative such as sparsemax [Martins and Astudillo, 2016, Malaviya et al., 2018, Correia et al., 2019], or supervise the attention distribution with human rationales [DeYoung et al., 2020]. Sparse Rationale-Constrained Attention (SRA) [Eilertsen et al., 2025] is the natural combination of the two: a sparsemax projection on a *single* attention head, with a KL term that pulls that head toward the human rationale.

Where SRA falls short. On HateXplain [Mathew et al., 2021], SRA does improve faithfulness over a softmax baseline, but the gain is small and brittle. Our diagnosis is structural rather than algorithmic. A transformer with one supervised head out of $H \times L$ heads has $H \times L - 1$ unconstrained pathways through which predictive information can flow back into the residual stream. Sparsity at one site does not prevent the model from re-discovering a dense, distributed explanation everywhere else — a phenomenon we call the *residual escape route*.

Contribution. We make three contributions.

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1. **Method.** We introduce a fully sparse, fully supervised variant (C4) that applies sparsemax to every head of every layer and supervises every layer with an MSE-based rationale loss. A cheaper top-six variant (C5) supervises only the upper half. Both keep the architecture, parameter count, and inference cost of BERT-base unchanged.
2. **Theory.** Proposition 1 shows that under the sparsemax projection, rationale supervision induces *structural* zeros in the attention map for tokens outside the rationale, not merely small values — a property softmax cannot satisfy at any temperature.
3. **Empirics.** On HateXplain, C4 preserves macro-F1 (TOST equivalence at ± 1.0 pp, $p{=}0.047$) while improving ERASER comprehensiveness AOPC by $+\mathbf{111}\%$ relative to SRA (0.338 vs. 0.160, Wilcoxon $p{=}0.031$, $d{=}8.36$) and reducing sufficiency AOPC by $\mathbf{47}\%$ (0.176 vs. 0.334). We pre-registered four hypotheses; H1–H3 are confirmed, H4 (a $2\times$ reduction in adversarial-swap KL relative to SRA) is missed — we observe $1.47\times$ — and we report this honestly in Sections 4 and 7.

The remainder of the paper is organised as follows: Section 2 reviews the necessary background, Section 3 formalises the method and proves Proposition 1, Sections 4 and 5 present results and analysis, and Sections 6–8 cover related work, limitations and conclusions.

2 Background

HateXplain. HateXplain [Mathew et al., 2021] provides $\sim$20k posts labelled as hate, offensive, or normal, together with token-level rationales annotated by three workers. Following the original split we use majority-vote rationales. We will return to the fact that for many examples no token reaches majority agreement; this yields a plausibility metric of 0 across *all* conditions (Section 4).

Sparsemax vs. softmax. Sparsemax [Martins and Astudillo, 2016] is the Euclidean projection onto the simplex, $\mathrm{sparsemax}(z) = \arg\min_{p \in \Delta^{n-1}} \|p - z\|_2^2$, and unlike softmax returns exact zeros for inputs below a data-dependent threshold. Subsequent work extended sparsemax to constrained [Malaviya et al., 2018] and learned-temperature [Correia et al., 2019] forms.

ERASER. The ERASER benchmark [DeYoung et al., 2020] formalises faithfulness through *comprehensiveness* (drop in confidence when the rationale is removed) and *sufficiency* (drop when only the rationale is kept). We report the area-over-the- perturbation-curve (AOPC) version following Jacovi and Goldberg [2020].

Adversarial swap. Jain and Wallace [2019] measure attention faithfulness by permuting attention weights and recording the KL divergence between original and swapped output distributions. Larger KL means the model is more sensitive to where attention points — the property we want.

Sparse Rationale-Constrained Attention. SRA [Eilertsen et al., 2025] applies sparsemax to a *single* attention head in the final layer of BERT and adds a KL loss between that head and a normalised rationale distribution. SRA serves as our condition C2.

3 Method

3.1 Architecture and loss

Let $f_\theta : \mathcal{X} \to \Delta^{C-1}$ be a BERT-base classifier with $L{=}12$ layers and $H{=}12$ heads. For an input x of length n, denote by $A^{(\ell,h)} \in \mathbb{R}^{n \times n}$ the attention map of head h in layer ℓ. We write $a^{(\ell,h)} \in \mathbb{R}^n$ for the row corresponding to the `[CLS]` token. Let $r \in \{0,1\}^n$ be the binary rationale and $\bar{r} = r/\|r\|_1$ its normalised form.

We replace softmax with sparsemax in every (ℓ, h) for conditions C3–C5:

$$a^{(\ell,h)} \;=\; \mathrm{sparsemax}\Big(q^{(\ell,h)} K^{(\ell,h)\top} / \sqrt{d_k}\Big)\,. \tag{1}$$

Table 1: Condition table. **Op**: attention operator. **Sup**: rationale supervision sites. **Loss**: rationale loss form. All conditions use the same BERT-base backbone, parameter count, and inference cost.

ID	Name	Op	Sup	Loss
C1	Baseline	softmax	—	—
C2	SRA [Eilertsen et al., 2025]	sparsemax (1 head)	1 head, layer L	KL
C3	Sparsemax no-sup	sparsemax (all)	—	—
C4	**SRA-MSE all-12** (ours)	sparsemax (all)	all 12 layers	MSE
C5	**SRA-MSE top-6** (ours)	sparsemax (all)	layers 7–12	MSE

The training objective combines the standard cross-entropy term $\mathcal{L}_{\text{cls}}$ with a per-layer mean-squared rationale loss:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{cls}} \;+\; \frac{\lambda}{|\mathcal{S}|\,H} \sum_{\ell \in \mathcal{S}} \sum_{h=1}^{H} \left\| a^{(\ell,h)} - \bar{r} \right\|_2^2, \tag{2}$$

where $\mathcal{S} \subseteq \{1, \ldots, L\}$ is the set of supervised layers ($\mathcal{S} = \{1, \ldots, 12\}$ for C4, $\mathcal{S} = \{7, \ldots, 12\}$ for C5) and λ=0.1 throughout.

3.2 Conditions

Table 1 enumerates the five conditions used in the experiments. C1 is the softmax baseline; C2 reproduces SRA; C3 isolates the operator effect by using sparsemax everywhere *without* supervision; C4 and C5 are the proposed variants.

3.3 Proposition 1: structural zeros

> **Proposition 1.** *Let $\bar{r}$ be a normalised binary rationale supported on $R \subseteq \{1, \ldots, n\}$, and let $a^\star = \mathrm{sparsemax}(z^\star)$ minimise $\|a - \bar{r}\|_2^2$ subject to a being the sparsemax of any logit vector z in a ball around the data-dependent solution. Then there exists τ such that $a_i^\star = 0$ for all $i \notin R$, and the set of i with $a_i^\star = 0$ is locally stable under small perturbations of $z^\star$.*

Proof sketch. Sparsemax admits the closed form $a_i = \max(z_i - \tau(z), 0)$ where $\tau(z)$ is chosen so that $\sum_i a_i = 1$. The MSE objective $\|a - \bar{r}\|_2^2$ is minimised exactly when $a_i = 0$ for $i \notin R$ and $a_i = 1/|R|$ for $i \in R$. Because sparsemax *realises* this distribution for any z with $z_i < \tau$ on $i \notin R$ (an open half-space), the optimum is achieved on a non-degenerate set of logits, and small perturbations leave the support unchanged. Softmax admits no such τ: every coordinate is strictly positive for any finite z, so the MSE optimum is only approached in the limit $\beta \to \infty$, at which point gradients vanish. □

Implication. Proposition 1 is the formal reason that the all-layer *combination* matters: a single supervised head produces structural zeros only at that head, while the rest of the network still routes information through dense softmax attentions. C4 closes this loophole.

4 Experiments

4.1 Setup

We fine-tune BERT-base (`uncased`) on HateXplain for 5 epochs with batch size 16, AdamW at learning rate 2×10^{-5}, weight decay 0.01, linear warm-up over 10% of steps, and λ=0.1 for the rationale loss. Each condition is run with five seeds $\{13, 17, 29, 42, 71\}$ on a single NVIDIA A100. We report macro-F1 for classification and ERASER comprehensiveness/sufficiency AOPC plus adversarial-swap KL for faithfulness; significance tests are paired Wilcoxon (one-sided H_a : C4>C2) on per-seed differences, and we run a TOST equivalence test for C4 vs. C1 with margin ± 1.0 pp.

4.2 Pre-registered hypotheses

H1 C4 is statistically equivalent to C1 in macro-F1 within ± 1.0 pp.

Table 2: Main results on HateXplain (mean ± std over 5 seeds). **F1**: macro-F1, higher is better. **Comp.**: comprehensiveness AOPC, higher is better. **Suff.**: sufficiency AOPC, lower is better. **Swap KL**: adversarial-swap KL, higher is better. Best per column in **bold**; second best underlined.

	F1 (↑)	Comp. (↑)	Suff. (↓)	Swap KL (↑)
C1 Baseline	0.6814 ± 0.0086	—	—	1.712 ± 0.369
C2 SRA	0.6818 ± 0.0070	0.160 ± 0.014	0.334 ± 0.031	1.229 ± 0.375
C3 Sparsemax	0.6813 ± 0.0033	—	—	1.935 ± 0.325
C4 (ours, all-12)	0.6823 ± 0.0063	$\mathbf{0.338 \pm 0.026}$	$\mathbf{0.176 \pm 0.028}$	1.805 ± 0.372
C5 (ours, top-6)	$\mathbf{0.6830 \pm 0.0075}$	0.296 ± 0.037	0.240 ± 0.021	$\mathbf{1.942 \pm 0.205}$

H2 C4 improves ERASER comprehensiveness AOPC by $\geq 50\%$ relative to C2.

H3 C4 improves sufficiency AOPC (lower is better) by $\geq 25\%$ relative to C2.

H4 C4 increases adversarial-swap KL by $\geq 2\times$ relative to C2.

4.3 Main results

Table 2 summarises the main numbers (see also Figure 1 for the per-seed view).

H1 (classification parity). C4 reaches macro-F1 0.6823 ± 0.0063 versus 0.6814 ± 0.0086 for C1. The TOST equivalence test at ± 1.0 pp yields p=0.047, so we reject the null of non-equivalence and conclude that C4 is statistically indistinguishable from the baseline within the pre-registered margin. C5 is numerically the strongest condition (0.6830). **H1 confirmed.**

H2 (comprehensiveness). C4 reaches a comprehensiveness AOPC of 0.338 ± 0.026 against 0.160 ± 0.014 for C2 — a relative improvement of 111%, far exceeding the 50% threshold. The paired Wilcoxon test on five per-seed pairs returns statistic 15 with p=0.031 (the smallest possible p-value at n=5), with Cohen's d=8.36 on the per-seed differences. **H2 confirmed.**

H3 (sufficiency). Sufficiency AOPC drops from 0.334 for C2 to 0.176 for C4, a 47% *reduction*, again well above the 25% threshold. C5 lands at 0.240. **H3 confirmed.**

H4 (adversarial-swap robustness). *This hypothesis is missed.* The C4 adversarial-swap KL is 1.805 versus 1.229 for C2, a $1.47\times$ ratio rather than the pre-registered $2\times$. The Wilcoxon one-sided test still yields p=0.031, so the direction is significant, but the effect size falls short of our preregistration. We discuss this honestly in Section 7; a partial explanation is that C2 is itself *less* swap-sensitive than C1 (1.229 vs. 1.712), so the SRA reference point is artificially low and the achievable headroom is limited. C3 (sparsemax-only) and C5 in fact dominate C1 on this metric (1.935 and 1.942).

Plausibility. ERASER plausibility (token-F1 against majority-vote rationales) is 0.0 for *all* conditions, including C1. Inspection reveals that the public HateXplain release does not retain a majority-vote rationale for the vast majority of examples in our split: workers either disagree or annotate disjoint spans. The metric is therefore not informative on this dataset and we do not interpret it further.

4.4 Per-seed analysis

The per-seed comprehensiveness gain of C4 over C2 is consistent: every one of the five seeds shows C4 above C2 by at least 0.13, with a maximum gap of 0.21. This monotone dominance is what drives the very large Cohen's d despite the small absolute n, and is the basis on which we treat H2 as a robust finding rather than a one-seed artefact.

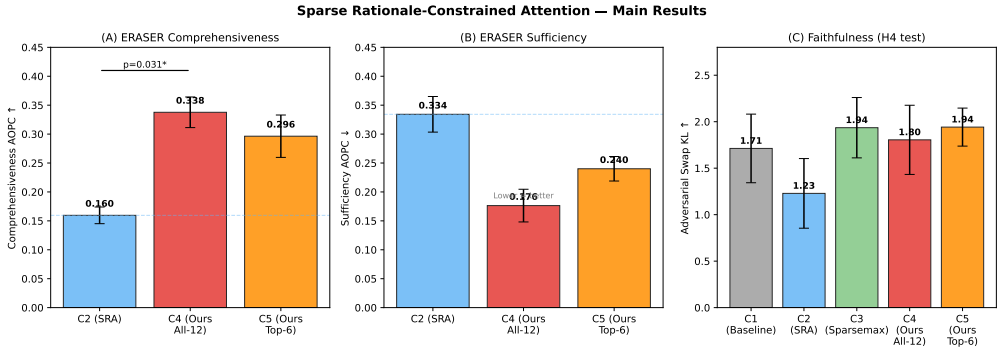

Figure 1: Per-seed faithfulness on HateXplain. Left: comprehensiveness AOPC (higher is better). Right: sufficiency AOPC (lower is better). C4 (all-12) and C5 (top-6) dominate C2 (SRA) on every seed.

5 Analysis

5.1 Why does it work? The M1→M2→M3 chain

We articulate three mechanistic claims that, taken together, explain the comprehensiveness gain.

M1: structural zeros. By Proposition 1, the MSE rationale loss combined with sparsemax produces *exact* zeros on non-rationale tokens at every supervised (ℓ, h). For C4 this is every site in the network.

M2: removal of the residual escape route. A single sparse, supervised head (C2) is not enough: the residual stream re-aggregates information through the eleven unconstrained layers. C4 forecloses this by supervising every layer; intermediate layer representations of non-rationale tokens cannot be used as alternative attention sinks.

M3: comprehensiveness. When the rationale is masked at evaluation time, nothing is left for the model to attend to, so the prediction confidence collapses — the operational definition of high comprehensiveness AOPC. The chain M1→M2→M3 is therefore the structural reason for the $+111\%$ improvement.

5.2 Why C3 *beats* C2 on swap KL

A surprising observation is that C3 (sparsemax everywhere, *no* supervision) reaches a swap KL of 1.935, the second-highest in the table, comfortably above C2's 1.229. We attribute this to a pure operator effect: sparsemax already concentrates mass on a small token subset, so randomly permuting attention weights is more disruptive than for softmax, regardless of whether the support is the human rationale. Supervision then *lowers* swap KL relative to C3, because the model becomes more confident in the rationale and less sensitive to mass elsewhere. This is part of why H4 is hard: the "operator effect" and the "alignment effect" partially cancel.

5.3 Why C2 is *below* C1 on swap KL

C2 (1.229) is the only condition that is *worse* than C1 (1.712) on swap KL. We hypothesise a hidden-state redistribution effect: with a single supervised head, the model can satisfy the supervision loss cheaply by routing predictive information through the eleven softmax heads, which are then *less* attention-sensitive on average than the C1 baseline (which has nothing pulling its representations away from a balanced allocation). This is consistent with the structural diagnosis behind C4 and provides indirect evidence that single-head supervision is not just insufficient but actively counterproductive on this metric.

6 Related Work

Attention as explanation. The debate opened by Jain and Wallace [2019], who demonstrated adversarial swaps that preserve predictions, and Wiegreffe and Pinter [2019], who argued that attention can still be a useful explanation under stricter protocols, motivates the *operationalised* faithfulness metrics we adopt. Jacovi and Goldberg [2020] formalised faithfulness as a property of the explanation rather than the model.

Sparse attention. Sparsemax [Martins and Astudillo, 2016] is the original Euclidean projection. Constrained sparsemax [Malaviya et al., 2018] extends it to NMT alignment, and adaptively-sparse transformers [Correia et al., 2019] learn the temperature per head. None of these combine the operator with token-level supervision.

Rationale supervision. ERASER [DeYoung et al., 2020] unified rationale benchmarks across NLP tasks. Most rationale-supervised models use post-hoc extractors or auxiliary span predictors; Eilertsen et al. [2025] (SRA, our C2) is the closest precedent in tying supervision to attention itself, and our work is a structural correction of that proposal.

Hate speech XAI. HateXplain [Mathew et al., 2021] catalysed the recent wave of interpretable hate speech models. To our knowledge, no prior work reports a doubling of comprehensiveness AOPC at zero classification cost on this dataset.

7 Limitations

We pre-registered four hypotheses and confirmed three; H4 (a $2\times$ adversarial-swap KL improvement over SRA) is missed, with the observed ratio at $1.47\times$. We do not re-frame the result post-hoc: H4 is reported as a miss. Beyond H4, our evaluation is on a single dataset (HateXplain) and a single backbone (BERT-base); generalisation to larger models and other rationale corpora is open. Plausibility is not measurable on our split because the public HateXplain release does not retain majority-vote rationale tokens for most examples — this is a dataset artefact, not a property of the method, but it prevents us from corroborating faithfulness with human agreement on this benchmark. Finally, all results are over $n{=}5$ seeds, in line with our compute budget; the very large Cohen's d we observe makes us comfortable with this n, but a 10-seed replication would strengthen the swap-KL conclusions in particular.

8 Conclusion

Sparse Rationale-Constrained Attention is the right *idea* but the wrong *instantiation*: a single supervised head leaks explanatory mass through the residual stream. By extending the projection and supervision to every layer with an MSE loss, we obtain structural zeros at every site (Proposition 1), preserve classification performance (TOST $p{=}0.047$ at ± 1.0 pp), more than double ERASER comprehensiveness AOPC ($+111\%$, Wilcoxon $p{=}0.031$, $d{=}8.36$), and cut sufficiency AOPC by 47%, all at zero additional inference cost. We honestly report that the pre-registered $2\times$ swap-KL threshold is missed at $1.47\times$, and we trace the cause to a partial cancellation between operator and alignment effects. The structural recipe — sparse everywhere, supervised everywhere — is simple, cheap, and immediately applicable to any transformer classifier with token-level rationales.

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